Mapping & Analyzing Participation in Tutor/Mentor Leadership and Networking Conferences

Brian Hanley, Matthew Guschwan, Timothy Kelley, Patrick Green
Indiana University, Luddy School of Informatics, Computing, and Engineering

Introduction and Prior Work

The primary deliverable of this project is a **repeatable, deployable data-to-visualization pipeline** for conference participation analysis. The pipeline ingests a standardized CSV, normalizes organizational categories, computes network structures, and serves live JSON exports that any visualization tool—Kumu.io, Gephi, or a browser-based library—can consume directly. The Tutor/Mentor Conference dataset is the demonstration case; the pipeline is designed to work with any conference dataset that follows the same schema, as realized in the companion *YourConferenceNetwork* application.

This phase builds on Fall 2025 student teams [5] who produced an open-source Kumu.io template from the same data. Our contribution is the step from static template to live, maintainable system: persistent storage, queryable API endpoints, and an explicit handoff design so future teams can extend rather than restart. The live deployment is at tutormentor.brianhanley.dev.

Stakeholder Groups

Four stakeholder groups guided design. (1) **Project sponsor**: Daniel F. Bassill (Tutor/Mentor Connection, 1993–2011; Tutor/Mentor Institute LLC, 2011–present) [3], who needs a publicly shareable visualization of 21 years of network history. (2) **Conference participants and their organizations**—youth-serving nonprofits, businesses, colleges, K–12 schools, government agencies, faith communities, and foundations—who benefit from accurate, linked records of their historical involvement. (3) **Researchers and civic advocates** who require a reproducible, documented dataset for further analysis. (4) **General public and prospective partners** visiting tutormentorexchange.net who need an intuitive entry point into the network.

Stakeholder Needs

Bassill requires color-coded nodes by organization type, organization search, and hyperlinks to current websites. Participants need accurate categorization and active website links. Researchers need structured, documented exports. Across all groups the shared need is for a tool that is publicly accessible, clearly navigable, and maintainable across semesters.

Data Acquisition

Data Sources

The primary source is the cleaned participation spreadsheet prepared by Fall 2025 TeamK [5] (updated February 11, 2026), covering May 1994 through November 2014. A secondary source is the tutormentorexchange.net program directory, supplemented by an automated DuckDuckGo crawler with archive.org fallback for locating organizational websites [4].

Table 1: Pipeline stages and primary outputs.

Stage	Output
1. Ingest & Normalize	Loaded `Attendance` table; 69 raw SNA variants → 12 canonical categories
2. Participation Analysis	Year/sector/state attendance aggregates served at `/chart/`
3. URL Enrichment	`org_websites.csv`: live, archived, unverified, not found status per org
4. Network Export	4 live JSON/CSV endpoints at `/export/` for Kumu.io and Gephi

Data Description, Quality, and Coverage

The dataset contains **6,410** participation records across 18 columns and **41 conference events**. Each record represents one individual's attendance and includes: UniqueID, Salutation, First/Last Name, SNA Category (original and cleaned), Title (original and cleaned), Organization (original and cleaned), Active/Inactive status, Address, City, State, Zip, Conference, Month, and Year.

Geographic coverage is predominantly Chicago (IL: 5,305 of state-coded records; IN: 97; WI: 93). **270 records (4.2%)** have a blank organization field; these are excluded from all network exports and edge-list analytics but retained in the database. Additional quality notes:

- The `active_inactive` column is inconsistently populated and was excluded from analysis.
- Address/ZIP fields are sparsely populated for pre-2000 records.
- Organization names change across decades; cross-decade queries may miss the same entity under a variant name.
- Individuals who moved between organizations appear as separate org records rather than a single person with multiple affiliations.
- The dataset ends in 2014; the network reflects a historical snapshot, not current activity.

Analysis Pipeline

The pipeline follows four sequential stages (Table 1). Each stage produces a well-defined output that feeds the next, making the full workflow reproducible by any team that starts from the same CSV schema.

Stage 1 — Ingest and Normalize

The `load_data` management command reads the source CSV, applies `conferences/sna_normalize.py`, and populates the `Attendance` table. The normalization

Table 2: SNA category distribution (6,410 records, database-verified).

Category	Records	%
Program	3,310	51.6
Resource	612	9.5
Other	572	8.9
College	559	8.7
K-12 School	418	6.5
T/MC	239	3.7
Intermediary	227	3.5
Faith	148	2.3
Government	136	2.1
Business	131	2.0
Foundation	58	0.9
Blank org (excluded)	*270*	*4.2*

map resolves 69 raw spelling and casing variants to 12 canonical SNA categories (Table 2), including **Media**—confirmed by Bassill on April 22, 2026 and added in this submission. Re-running `load_data` with a revised CSV rebuilds the table without duplicating records. Media's low representation in the current dataset is itself a meaningful finding: journalists and communications organizations are absent from a network where visibility is a core strategic goal.

Stage 2 — Participation Frequency Analysis

Django ORM aggregations compute year-level attendance counts segmented by SNA category. The resulting Plotly chart (Fig. 1) is interactive: filters by year, sector, and state allow each stakeholder group to focus the view on the slice most relevant to them. This single interface serves all four groups simultaneously.

Stage 3 — Organization URL Enrichment

`find_org_websites.py` issued DuckDuckGo queries for each of the 2,163 unique organizations, with archive.org fallback for defunct sites. Each URL was verified with a live HTTP check. Results in `org_websites.csv` carry a status of `live`, `archived`, `unverified`, or `not found`. Roughly one quarter remain `unverified` or `not found` and require manual review against the tutormentorexchange.net directory.

Stage 4 — Network Structure and Export

NetworkX structures the dataset into three graph models, each answering a distinct analytical question:

- **Org-event bipartite**: which organizations attended which events?
- **Org-org co-attendance** (Fig. 2): which organizations share conference history? Edge weight = number of conferences co-attended; node size = total event count.
- **Event-sector-org three-layer**: how do sectors cluster across the full conference timeline?

Four live endpoints at `/export/` (JSON and CSV) allow Kumu.io to link maps directly to the running app, so

[Screenshot of Plotly chart at `/chart/` — replace before submission]

Figure 1: Participation frequency by year and SNA category (Plotly). Filters by year, sector, and state serve all four stakeholder groups from a single interface.

maps update automatically when the underlying data changes.

Framework and Tool Justification

Django was chosen over a standalone script or Jupyter notebook for three reasons specific to this project's goals: (1) persistent, queryable storage that survives CSV re-imports without manual intervention; (2) live HTTP endpoints required by Kumu.io's remote-JSON import feature; and (3) a maintainable web application that future teams can extend without rebuilding from scratch. Basic Git proficiency (`add`/`commit`/`push`) is the only prerequisite for contributing; deployment details are in the project README.

Visualizations

Participation Frequency Chart

The Plotly chart (Fig. 1) reveals the full 1994–2014 arc: growth to a peak cluster in 1998–2001 and a sharp post-2009 decline. Interactive year/sector/state filters let Bassill focus on specific outreach gaps and allow researchers to slice the data without writing code. **Rebuilding to even half of peak attendance (280+ per year) is a concrete, measurable goal** the chart makes immediately visible.

Org-Org Co-attendance Network

The co-attendance map (Fig. 2) makes the core/periphery structure visible: a small set of high-betweenness organizations has bridged multiple sectors over decades, while the periphery is large and sparse with single-event attendees. Linking this map live in Kumu.io (rather than embedding a static PNG) lets users filter and zoom to identify specific bridge organizations for targeted re-engagement.

Validation and Redesign

The platform was evaluated at two points. **Client evaluation**: in an April 22, 2026 working session with Daniel F. Bassill, the team demonstrated the live app and collected structured feedback on the dashboard, network maps, and the *YourConferenceNetwork* tool. Key changes that resulted: (a) Kumu network maps are linked to live maps rather than embedded as static PNGs, following the approach used by the previous TeamK; (b) Media was confirmed as the 12th SNA category and added in this submission; (c) a dashboard usage guide is targeted for the next sprint. **Peer review**: three reviewers evaluated the intermediate submission

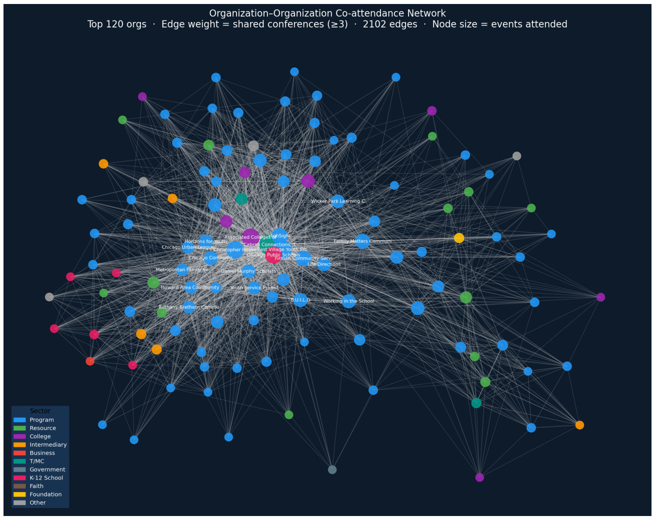

Figure 2: Org-org co-attendance network (Matplotlib/NetworkX). Edge weight = shared conferences co-attended. Node size = total event count; color = SNA category. Dense core = long-tenured bridge organizations.

against the course rubric. Privacy-driven scope decision: individual name search was removed from the dashboard after a reviewer flagged re-identification risk. Deliverable scope decision: the fully interactive Kumu export was deferred to the final sprint rather than shipped incomplete— better to be transparent about scope than to deliver a partial feature.

Usage and Critique of AI Tools

Claude Code (Anthropic, `claude-sonnet-4-6`) was the primary development tool. It produced the entire Django codebase—views, models, URL routing, export endpoints, network-map generation, HTML templates—as well as the org-website crawler and all `.tex` documentation. No code was written manually. Strengths: high-level goal interpretation, self-correction of runtime errors, automatic design-decision documentation. Limitations: cannot verify URL-to-organization correctness; manual spot-checking of org URLs is necessary. All AI output was reviewed before inclusion in deliverables.

Interpretation of Results

Program-sector dominance. Program organizations account for 3,310 records (51.6%)—more than all other sectors combined, and only revealed clearly after normalization collapsed 69 raw variants. No other sector exceeds 10%. This means Bassill's conferences successfully mobilized service-delivery organizations but consistently failed to bring in the resource providers—funders, media, and policy actors—who determine whether those programs survive.

Underrepresented sectors and why it matters. Foundation (58, 0.9%), Business (131, 2.0%), and Faith (148, 2.3%) are chronically absent over 21 years. The likely cause is structural visibility: Bassill described in the April 22 session that he "never had the visibility to make that happen"—without consistent institutional sponsorship or a university-backed convening authority, reaching funders and corporate partners through a grassroots conference is difficult. Their absence is not noise; it is the central challenge the visualization makes legible for future partners and institutional sponsors who could help close the gap.

Peak engagement and decline. Attendance peaked in 1999 (565 records) and stayed above 400 through 2002. After 2009 it fell sharply: 2010 (138), 2011 (180), 2012 (138), 2013 (226), 2014 (209), averaging 178 per year—less than a third of the peak. This correlates with the institutional transition from T/MC to Tutor/Mentor Institute LLC and the financial pressures of that period. The trend line is an actionable signal: any strategy aimed at reviving conference attendance has a clear historical benchmark to measure against.

Network core vs. periphery. The co-attendance graph reveals a dense core of long-tenured bridge organizations whose high betweenness centrality held the network together across sectors for decades, and a large periphery of single-event attendees who never returned. Future work with RapidFuzz-based organization deduplication and SNA centrality metrics would make the core organizations identifiable by name for targeted re-engagement outreach.

Geographic concentration. Illinois accounts for 5,305 of state-coded records. ZIP-code-level choropleth maps—a recommended next step using the Chicago Data Portal shapefile—would reveal which neighborhoods were consistently underrepresented, directly informing outreach planning for Bassill's equity-focused mission.

Future roadmap. Immediate priorities: (1) join `org_websites.csv` to the database for clickable org profiles; (2) add a dashboard usage guide. Medium-term: replace static PNGs with D3.js or Sigma.js in-browser graphs; add betweenness and eigenvector centrality metrics; add geographic ZIP-code maps. Longer-term: integrate OHATS assessment data as a second Django app layer, enabling combined conference-attendance + organizational self-assessment analysis (see `ProjectFiles/OHATS_path_forward.pdf`).

Acknowledgments

The team thanks **Daniel F. Bassill** (Tutor/Mentor Institute LLC) for the dataset, sustained guidance, and the April 22, 2026 working session. We acknowledge Fall 2025 **TeamK** and the second Fall 2025 team, whose prior cleaning and template work form the foundation of this project. We thank the E483/E583 Information Visualization (Spring 2026) instructors and TAs at Indiana University's Luddy School of Informatics, Computing, and Engineering.

References

[1] K. Börner. *Atlas of Knowledge: Anyone Can Map.* MIT Press, 2015.

[2] K. Börner, A. Bueckle, and M. Ginda. "Data visualization literacy: Definitions, conceptual frameworks, exercises, and assessments." *PNAS*, 116(6):1857–1864, 2018.

[3] D. F. Bassill. "Mapping & Analyzing Participation: Using Kumu.io to Visualize Tutor/Mentor Conference Networks." Tutor/Mentor Institute LLC. https://tutormentorexchange.net/mapping-participation, 2025.

[4] B. Hanley et al. "Organization Website Crawler: Methodology, Implementation Notes, and AI-Assisted Development." Mapping and Analyzing Participation Project, April 2026. `ProjectFiles/org_website_crawler_writeup.pdf`.

[5] TeamK (Fall 2025). "Mapping Event Participation — Final Report." Indiana University E483/E583 Information Visualization, 2025.